

Syllabus:- COMPLEX NUMBER

SECTION - I

Multiple Choice Questions (MCQ)

This section contains **20 multiple choice** questions. Each question has 4 choices (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** is correct.

- If z_1, z_2, z_3 are distinct complex numbers such that $\frac{3}{|z_2 - z_3|} = \frac{4}{|z_3 - z_1|} = \frac{5}{|z_1 - z_2|}$, then the value of $\frac{9}{z_2 - z_3} + \frac{16}{z_3 - z_1} + \frac{25}{z_1 - z_2}$ equals
 (1) 0 (2) 3 (3) 4 (4) 5
- The locus of complex number $z = x + iy$ where $i = \sqrt{-1}$ satisfying relation $\arg(z - a) = \frac{\pi}{4}$ where $a \in \mathbb{R}$ is
 (1) $x^2 - y^2 = a^2$ (2) $x^2 + y^2 = a^2$ (3) $x + y = a, y > 0$ (4) $x - y = a, y > 0$
- Let $\bar{z}$ be complex conjugate of z and $k \in \mathbb{R}$, then the curve represented by $\operatorname{Re}(\bar{z})^2 = k^2$ can be a parabola, ellipse, hyperbola or a pair of straight lines. How many of these curves are possible?
 (1) 1 (2) 2 (3) 3 (4) 4
- If $x^2 - x + 1 = 0$, then $\sum_{n=1}^5 \left(x^n + \frac{1}{x^n} \right)^2 =$
 (1) 8 (2) 5 (3) 12 (4) 20
- The real part of $(\sin x + i \cos x)^5$, where $i = \sqrt{-1}$ is
 (1) $-\cos 5x$ (2) $-\sin 5x$ (3) $\cos 5x$ (4) $\sin 5x$
- If $\omega = \sqrt{-1}$, then, $\left(\frac{i + \sqrt{3}}{-i + \sqrt{3}} \right)^{52722} + \left(\frac{i - \sqrt{3}}{i + \sqrt{3}} \right)^{40305} =$
 (1) $\sqrt{2}$ (2) $\sqrt{3}$ (3) 2 (4) 3
- Number of imaginary roots of equation $(2x - 3)^4 = 81$ are
 (1) 0 (2) 1 (3) 2 (4) 4
- If $z = x + iy, w = \frac{1 - iz}{z - i}, |w| = 1$ implies that in complex plane, z lies on the
 (1) imaginary axis (2) real axis (3) unit circle (4) parabola
- If z is a complex number of constant modulus such that z^2 is purely imaginary then the number of possible values of z is
 (1) 2 (2) 1 (3) 4 (4) infinite
- If ω is a non real cube root of unity then $\frac{1 + 2\omega + 3\omega^2}{2 + 3\omega + \omega^2} + \frac{2 + 3\omega + \omega^2}{3 + \omega + 2\omega^2} =$
 (1) -1 (2) 2ω (3) 0 (4) -2ω

11. If $z_r = \cos\left(\frac{2r\pi}{5}\right) + i\sin\left(\frac{2r\pi}{5}\right)$ where $r = 0, 1, 2, 3, 4$ then $z_1 z_2 z_3 z_4 z_5 =$
- (1) -1 (2) 0 (3) 1 (4) none of these
12. If $\arg\left(\frac{z-1}{z+1}\right) = \frac{\pi}{3}$, then locus of z is
- (1) square (2) arc of circle (3) pair of lines (4) branch of hyperbola
13. Let w ($\text{Im}(w) \neq 0$) be a complex number. Then the set of all complex numbers z satisfying the equation $w - \bar{w}z = k(1 - z)$, for some real number k , is
- (1) $\{z : |z| = 1\}$ (2) $\{z : z = \bar{z}\}$ (3) $\{z : z \neq 1\}$ (4) $\{z : |z| = 1, z \neq 1\}$
14. If z_1, z_2 and z_3, z_4 are two pairs of complex conjugate numbers, then $\arg\left(\frac{z_1}{z_4}\right) + \arg\left(\frac{z_2}{z_3}\right)$ can be
- (1) π (2) $\frac{\pi}{2}$ (3) $\frac{3\pi}{2}$ (4) 0
15. Let $z = 1 + ai$ be a complex number, $a > 0$ such that z^3 is a real number. Then the sum $1 + z + z^2 + \dots + z^{11}$ is equal to
- (1) $-1250\sqrt{3}i$ (2) $1250\sqrt{3}i$ (3) $-1365\sqrt{3}i$ (4) $1365\sqrt{3}i$
16. Let $z \in \mathbb{C}$, the set of complex numbers. Then the equation, $2|z + 3i| - |z - i| = 0$ represents :
- (1) a circle with diameter $\frac{10}{3}$. (2) a circle with radius $\frac{8}{3}$
- (3) an ellipse with length of major axis $\frac{16}{3}$. (4) an ellipse with length of minor axis $\frac{16}{9}$
17. Two points P and Q in the Argand diagram represent complex numbers z and $3z + 2 + i$. If P moves around the circle with centre at origin and radius 2, then Q moves on the circle, whose centre and radius are
- (1) $-2 - i, 6$ (2) $2 - i, 3$ (3) $2 + i, 6$ (4) $2 + i, 3$
18. If $|z_1| = |z_2| = |z_3| = 1$ and $z_1 + z_2 + z_3 = \sqrt{2} + i$, then the number $z_1\bar{z}_2 + z_2\bar{z}_3 + z_3\bar{z}_1$ is:
- (1) a positive real number (2) a negative real number
- (3) always zero (4) a purely imaginary number
19. The set of all $\alpha \in \mathbb{R}$, for which $w = \frac{1+(1-8\alpha)z}{1-z}$ is a purely imaginary number, for all $z \in \mathbb{C}$ satisfying $|z| = 1$ and $\text{Re } z \neq 1$, is :
- (1) $\{0\}$ (2) $\left\{0, \frac{1}{4}, -\frac{1}{4}\right\}$ (3) equal to $\mathbb{R}$ (4) an empty set
20. If the complex number z satisfies the condition $|z| \geq 3$, then the least value of $\left|z + \frac{1}{z}\right|$ is equal to
- (1) $5/3$ (2) $8/3$ (3) $11/3$ (4) $13/3$

SECTION-II
(NUMERICAL VALUE TYPE QUESTIONS)

This section contains Five (5) questions. The answer to each question is **NUMERICAL VALUE** with twodigit integer and decimal upto one digit.

21. Find the number of complex number satisfying equation $z^3 = \bar{z}$ and $\arg(z + 1) = \frac{\pi}{4}$ simultaneously
22. Let z and w be non-zero complex numbers such that $zw = |z|^2$ and $|z - \bar{z}| + |w + \bar{w}| = 4$. If w varies then find the area enclosed by locus of z .
23. The radius of circle $z\bar{z} + (4 - 3i)z + (4 + 3i)\bar{z} + 5 = 0$ is r then $\frac{r}{\sqrt{5}}$
24. The largest value of r for which the region represented by the set $\{\omega \in \mathbb{C} \mid |\omega - 4 - i| \leq r\}$ is contained in the region represented by the set $\{z \in \mathbb{C} \mid |z - 1| \leq |z + i|\}$ is equal to k then find value of $\sqrt{2} \times k$
25. The equation $\operatorname{Im}\left(\frac{iz - 2}{z - i}\right) + 1 = 0$, $z \in \mathbb{C}$, $z \neq i$ represent a part of a circle. If the radius of the circle is r then find value of $4r$

Answer Key_DPP-22

1.	(1)	2.	(4)	3.	(2)	4.	(1)	5.	(4)	6.	(3)	7.	(3)
8.	(2)	9.	(3)	10.	(2)	11.	(3)	12.	(2)	13.	(4)	14.	(4)
15.	(3)	16.	(2)	17.	(3)	18.	(4)	19.	(1)	20.	(2)	21.	(1)
22.	(8)	23.	(2)	24.	(5)	25.	(3)						